NPTEL Online Certification Course
Applied Accelerated Artificial Architecture
Assignment 8: Detailed Solution
Indian Institute of Technology Guwahati

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Question 1: According to the linear scaling rule, if the baseline learning rate is 0.075 at a batch size of 512, what should the new learning rate be if the batch size is increased to 4096?

- 0.2
- 0.6
- 0.75
- 0.8

Correct Answer: 0.6

Explanation: The linear scaling rule scales the learning rate proportionally with the batch size increase (4096/512=8). Multiplying the baseline learning rate of 0.075 by 8 yields 0.6.

Question 2: What is defined as the point past which increasing the batch size further stops providing proportional returns in training efficiency?

- Effective batch size
- Baseline batch size
- Divergence batch size
- Critical batch size

Correct Answer: Critical batch size

Explanation: The critical batch size marks the inflection point where hardware scaling efficiency drops and further batch size increases yield diminishing returns in gradient noise reduction.

Question 3: What happens during a training "warm-up" phase to prevent early divergence?

- The learning rate is immediately set to its maximum value.
- The learning rate is linearly ramped up from near zero to the full value.
- The batch size is gradually decreased over several iterations.
- The gradients are completely cleared for the first 500 steps.

Correct Answer: The learning rate is linearly ramped up from near zero to the full value.

Explanation: Starting with a very small learning rate and gradually increasing it prevents large, destabilizing gradient updates when the model parameters are still random and unstable.

Question 4: Why does a single slow GPU slow down the entire multi-GPU training process?

- Because all GPUs must finish a step and synchronize before moving to the next.
- Because the system automatically reduces the overall batch size.
- Because the slow GPU forces a roll back to the last checkpoint.
- Because it triggers an automatic switch from Adam to the LAMB optimizer.

Correct Answer: Because all GPUs must finish a step and synchronize before moving to the next.

Explanation: Data-parallel synchronous training relies on barrier synchronization at each iteration, meaning fast GPUs must idle and wait for the slowest worker to finish its gradient computation.

Question 5: Which term describes a system's ability to keep working or recover properly when a part of it fails?

- Sharding
- Elastic training
- Fault tolerance
- Rendezvous

Correct Answer: Fault tolerance

Explanation: Fault tolerance mechanisms ensure that distributed systems can detect node failures, preserve progress via checkpoints, and recover without restarting from scratch.

Question 6: If a training script saves a checkpoint at step 700 and a worker's GPU crashes at step 743, what happens when torchrun restarts the job?

- Training will start completely over from step 0.
- Training skips ahead to step 800 to avoid the corrupted block.
- Training continues exactly from step 743 without losing any work.
- Training resumes from step 700, losing 43 steps of work.

Correct Answer: Training resumes from step 700, losing 43 steps of work.

Explanation: Since checkpoints are saved periodically at discrete intervals, restarting after a crash rolls back execution to the most recent successfully saved state.

Question 7: What is a major advantage of utilizing an asynchronous save (async checkpoint) over a synchronous save?

- It splits the dataset into smaller, easier-to-manage batches.

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- It automatically reduces the learning rate to prevent memory fragmentation.
- It requires no background processing or coordination among workers.
- It hands the save to a background process so foreground training keeps running.

Correct Answer: It hands the save to a background process so foreground training keeps running.

Explanation: Asynchronous checkpointing serializes model weights in a separate background thread or process, preventing the main GPU training loop from stalling during disk I/O.

Question 8: Why is it highly discouraged to ship raw optimizer states along with a model for inference?

- It increases model size by 4x to 5x without aiding inference
- It restricts the model to cross-platform frameworks
- It prevents the framework from utilizing numeric formats
- It completely stops the training progression

Correct Answer: It increases model size by 4x to 5x without aiding inference

Explanation: Optimizer states like momentum and variance buffers are strictly required for backpropagation updates and gradient optimization, serving zero purpose during forward-pass inference.

Question 9: In a model registry, what status is given to a model version that is currently live and serving real user traffic?

- Staging
- Production
- Archived
- None

Correct Answer: Production

Explanation: The production tag designates the verified model artifact currently deployed in live operational environments to handle real client queries.

Question 10: Which deployment strategy involves sending a small slice of traffic to a new model version before fully switching over?

- Rolling update
- Blue-green deployment
- Canary deployment
- Shadow rollout

Correct Answer: Canary deployment

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Explanation: Canary deployments route a fractional percentage of live production traffic to a new model version to test stability and real-world behavior safely.

Question 11: In Kubernetes, what mechanism is used to check the health and status of a container?

- Rollback
- Artifact
- Probe
- Budget

Correct Answer: Probe

Explanation: Kubernetes uses liveness, readiness, and startup probes to periodically assess container health and determine if traffic should be routed to it.

Question 12: What is the default Kubernetes rollout strategy for most stateless services?

- Blue-green deployment
- Rolling update
- Canary deployment
- Shadow rollout

Correct Answer: Rolling update

Explanation: Rolling updates incrementally replace old pods with new pods one by one or in small batches, ensuring zero downtime during application deployments.

Question 13: What occurs during a linear learning rate warm-up over the first 10% of training steps?

- The learning rate increases gradually to its peak value.
- The learning rate drops instantly to zero.
- The batch size decreases at a steady rate.
- The model updates stop entirely for those steps.

Correct Answer: The learning rate increases gradually to its peak value.

Explanation: Warm-up safely scales the learning rate from zero up to the target learning rate, preventing explosive gradient updates at the start of training.

Question 14: Why does large batch training typically underperform if a high learning rate is immediately used without a warm-up period?

- It experiences too many optimization steps per epoch.

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- It suffers from fewer optimization steps per epoch.
- The dataset size automatically shrinks.
- The GPU memory completely runs out.

Correct Answer: It suffers from fewer optimization steps per epoch.

Explanation: Large batches lead to fewer total weight updates per epoch, meaning high initial learning rates can cause unstable gradient trajectories before the optimizer stabilizes.

Question 15: Which specific deployment target is the TorchScript (.ts) model packaging format typically used for?

- Hugging Face Transformers
- ONNX Runtime engines
- Cloud-hosted TensorRT servers
- Mobile or standalone C++ applications

Correct Answer: Mobile or standalone C++ applications

Explanation: TorchScript serializes PyTorch models into a static graph format, enabling execution in non-Python runtime environments like mobile apps and C++ backends.

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